

CSA17 – Artificial Intelligence: Assessment Tool 1 (Solutions)

COURSE: Artificial Intelligence (CSA17) — Assessment Tool 1 CO3: Knowledge Representation and Reasoning (Propositional Logic, FOL, Inference, Resolution, Chaining) Total Marks: 40 (4 Case Studies × 10 marks each)

Note on mark split: The paper does not print a mark value against each sub-part (a)/(b)/(c), only the total per Assessment Tool (40 marks over 4 case studies = **10 marks per case study**). Based on the rubric weighting given (Understanding 25%, Application 25%, Analysis 20%, Solution Design 20%, Clarity 10%) and standard AI-course convention, each case study's 10 marks is typically split as: **(a) 3 marks — Representation/Setup, (b) 4 marks — Core Method/Derivation, (c) 3 marks — Analysis/Evaluation**. Solutions below are organized under this split — adjust weighting if your instructor specifies a different breakdown.

Case Study 1 — Smart Medical Diagnosis System (10 marks)

Facts: Fever = True, Cough = True, Breathlessness = True

(a) Propositional Representation — 3 marks

Let:

- F = Fever
- C = Cough
- R = Rash
- B = Breathlessness
- Flu = Possible Flu
- Mea = Possible Measles
- Pne = Possible Pneumonia

Rules:

- R1: $F \wedge C \rightarrow \text{Flu}$
- R2: $F \wedge R \rightarrow \text{Mea}$
- R3: $C \wedge B \rightarrow \text{Pne}$

Facts: F = True, C = True, B = True (R is unknown/not asserted)

(b) Forward Chaining — 4 marks

Step	Known facts	Rule fired	Condition check	New fact derived
1	F, C, B	R1: $F \wedge C \rightarrow$	$F=T, C=T$ ✓	Flu

Step	Known facts	Rule fired	Condition check	New fact derived
		Flu		
2	F, C, B, Flu	R2: $F \wedge R \rightarrow$ Mea	R unknown ✗	— (cannot fire)
3	F, C, B, Flu	R3: $C \wedge B \rightarrow$ Pne	$C=T, B=T$ ✓	Pne

Result: Forward chaining derives **Possible Flu** and **Possible Pneumonia**. R2 never fires because Rash is not known to be true.

(c) Backward Chaining — 3 marks

Goal: Pneumonia (Pne)

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Pne
├── Rule R3:  $C \wedge B \rightarrow$  Pne
│   ├── Sub-goal C (Cough) -> matches fact "Cough = True" ✓
│   └── Sub-goal B (Breathlessness) -> matches fact "Breathlessness
= True" ✓
└── Both sub-goals proved TRUE -> Pne is PROVED

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Since both leaves (C and B) reduce directly to known facts, the goal “**Patient A has Pneumonia**” is confirmed true.

Case Study 2 — Autonomous Traffic Management Agent (10 marks)

Rules (FOL):

- Rule 1: $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$
- Rule 2: $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$
- Rule 3: $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Unification with Rule 1 — 3 marks

Rule 1 antecedent literals: $\text{Vehicle}(x), \text{EmergencyType}(x)$

Unify	With fact	Substitution
$\text{Vehicle}(x)$	$\text{Vehicle}(\text{Ambulance})$	$\theta_1 = \{x/\text{Ambulance}\}$
$\text{EmergencyType}(x)$	$\text{EmergencyType}(\text{Ambulance})$	consistent with θ_1

MGU: $\theta = \{x / \text{Ambulance}\}$

Applying θ to Rule 1: $\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance}) \rightarrow \text{ClearPath}(\text{Ambulance})$ Both antecedents are true facts, so **ClearPath(Ambulance)** is inferred.

(b) Resolution Proof of Proceed(CarA) — 4 marks

Step 1 — Convert to CNF (clausal form):

- C1: $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$ (Rule 1)
- C2: $\neg \text{ClearPath}(y) \vee \text{GreenSignal}(y)$ (Rule 2, GreenSignal part)
- C3: $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$ (Rule 3)
- C4: $\text{Vehicle}(\text{Ambulance})$
- C5: $\text{EmergencyType}(\text{Ambulance})$
- C6: $\text{Vehicle}(\text{CarA})$
- C7: $\text{Behind}(\text{CarA}, \text{Ambulance})$
- C8: $\neg \text{Proceed}(\text{CarA})$ <- negated goal

Step 2 — Resolution steps:

1. C1 $\{x/\text{Ambulance}\} + \text{C4} \rightarrow \neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$
2. Result(1) + C5 $\rightarrow$ **ClearPath(Ambulance)**
3. C2 $\{y/\text{Ambulance}\} + \text{ClearPath}(\text{Ambulance}) \rightarrow$ **GreenSignal(Ambulance)**
4. C3 $\{x/\text{CarA}, y/\text{Ambulance}\} + \text{C8} (\neg \text{Proceed}(\text{CarA})) \rightarrow \neg \text{Vehicle}(\text{CarA}) \vee \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance})$
5. Result(4) + C6 $\rightarrow \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance})$
6. Result(5) + C7 $\rightarrow \neg \text{GreenSignal}(\text{Ambulance})$
7. Result(6) + GreenSignal(Ambulance) (from step 3) $\rightarrow \square$ (**empty clause**)

A contradiction (empty clause) is derived, so the negation of the goal is unsatisfiable with the KB $\rightarrow$ **Proceed(CarA) is proved TRUE.**

(c) Sufficiency for Two Simultaneous Emergency Vehicles — 3 marks

The current KB is **not sufficient**:

- It has no predicate to compare or arbitrate between two vehicles that both satisfy $\text{EmergencyType}(x)$ at the same junction — Rule 1 would derive $\text{ClearPath}(x)$ for *both*, and Rule 2 would then assign GreenSignal to both, which is physically unsafe/contradictory in a real intersection.
- **Missing elements needed:**
 1. A $\text{Priority}(x)$ or $\text{Severity}(x)$ predicate to rank emergency vehicles.
 2. A $\text{Conflict}(x,y)$ predicate identifying vehicles on intersecting paths.
 3. An arbitration rule, e.g. $\text{Vehicle}(x) \wedge \text{Vehicle}(y) \wedge \text{EmergencyType}(x) \wedge \text{EmergencyType}(y) \wedge \text{Conflict}(x,y) \wedge \text{Priority}(x) > \text{Priority}(y) \rightarrow \text{ClearPath}(x) \wedge \neg \text{ClearPath}(y)$.

- **Justification:** Logically, the KB as given is *consistent* but *incomplete* — it can derive ClearPath for each emergency vehicle independently without ever detecting the conflict between them, so an unsafe joint state is reachable without any rule ever being violated.
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Case Study 3 — Agricultural Expert Reasoning System (10 marks)

KB:

- P1: SoilDry $\rightarrow$ IrrigationNeeded
- P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
- P3: $\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk
- Facts: SoilDry = True, CropWheat = True

(a) Conversion to CNF — 3 marks

P1: SoilDry $\rightarrow$ IrrigationNeeded = $\neg$ SoilDry $\vee$ IrrigationNeeded (*already CNF*)

P2: (IrrigationNeeded $\wedge$ CropWheat) $\rightarrow$ ApplyDripMethod = $\neg$ (IrrigationNeeded $\wedge$ CropWheat) $\vee$ ApplyDripMethod (*eliminate $\rightarrow$*) = $(\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat) $\vee$ ApplyDripMethod (*De Morgan / push $\neg$ inward*) = **$\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod** (*CNF, single clause*)

P3: ($\neg$ ApplyDripMethod $\wedge$ CropWheat) $\rightarrow$ CropAtRisk = $\neg$ ($\neg$ ApplyDripMethod $\wedge$ CropWheat) $\vee$ CropAtRisk = (ApplyDripMethod $\vee$ $\neg$ CropWheat) $\vee$ CropAtRisk = **ApplyDripMethod $\vee$ $\neg$ CropWheat $\vee$ CropAtRisk** (*CNF*)

Facts (unit clauses): SoilDry ; CropWheat

(b) Resolution Refutation for ApplyDripMethod — 4 marks

Negate the goal: **$\neg$ ApplyDripMethod**

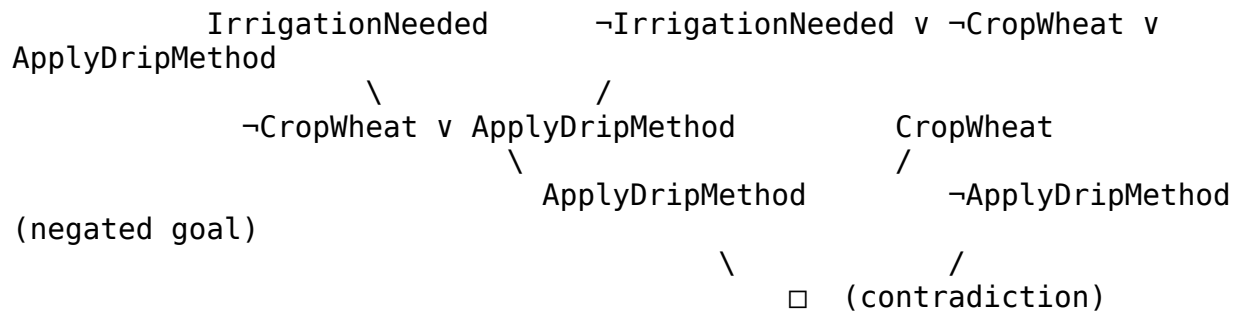
Clause set: C1 = $\neg$ SoilDry $\vee$ IrrigationNeeded, C2 = $\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod, C3 = SoilDry, C4 = CropWheat, C5 = $\neg$ ApplyDripMethod

1. C1 + C3 (SoilDry) $\rightarrow$ IrrigationNeeded
2. (1) + C2 $\rightarrow$ $\neg$ CropWheat $\vee$ ApplyDripMethod
3. (2) + C4 (CropWheat) $\rightarrow$ ApplyDripMethod
4. (3) + C5 ($\neg$ ApplyDripMethod) $\rightarrow$ $\square$ (empty clause)

The contradiction ($\square$) proves the theorem: **ApplyDripMethod holds** given the KB and facts.

Resolution proof tree:

SoilDry	¬SoilDry $\vee$ IrrigationNeeded
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(c) Effect of Removing “SoilDry” — 3 marks

- Without the fact SoilDry, **P1 cannot fire**, so **IrrigationNeeded can no longer be derived**.
 - Without IrrigationNeeded, **P2 cannot fire**, so **ApplyDripMethod can no longer be derived** (it becomes unknown, not proven false).
 - For P3 ($\neg\text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$) to fire, $\neg\text{ApplyDripMethod}$ must be *established*, not merely “not proven.” Two readings:
 - Under classical (open-world) logic:** ApplyDripMethod is simply *undetermined* — we cannot assert $\neg\text{ApplyDripMethod}$, so **CropAtRisk cannot be proven** by resolution refutation. The KB becomes under-constrained.
 - Under the Closed-World Assumption (CWA)**, commonly used in rule-based expert systems: anything not derivable is treated as false. Since ApplyDripMethod is not derivable, $\neg\text{ApplyDripMethod}$ is assumed true, P3 fires, and **CropAtRisk is concluded**.
 - Conclusion:** Whether “CropAtRisk still follows” depends on the reasoning assumption in use — it follows under CWA/negation-as-failure (typical of forward-chaining production systems), but does **not** follow under strict monotonic FOL resolution, since removing a fact can never *prove* a negative literal in classical logic.
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Case Study 4 — Wumpus World Knowledge Agent (10 marks)

Percepts: Stench[1,2], Breeze[1,1], Glitter[2,2]

Rules:

- Stench[1,2] $\rightarrow$ WumpusAdjacent[1,2]
- Breeze[1,1] $\rightarrow$ PitAdjacent[1,1]
- Glitter[x,y] $\rightarrow$ GoldAt[x,y]
- $\neg\text{Wumpus}[x,y] \wedge \neg\text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Belief State via Modus Ponens — 3 marks

Premise (percept, True)	Rule (Modus Ponens: $P, P \rightarrow Q \vdash Q$)	Conclusion derived
Stench[1,2]	Rule 1	WumpusAdjacent[1,2]

Premise (percept, True)	Rule (Modus Ponens: $P, P \rightarrow Q \vdash Q$)	Conclusion derived
Breeze[1,1]	Rule 2	PitAdjacent[1,1]
Glitter[2,2]	Rule 3	GoldAt[2,2]

Derived facts: WumpusAdjacent[1,2], PitAdjacent[1,1], GoldAt[2,2]

(b) Forward Chaining — Locating Wumpus and Pit — 4 marks

- **WumpusAdjacent[1,2]** $\Rightarrow$ Wumpus is in one of the neighbors of [1,2]: candidate set = {[1,1], [1,3], [2,2]}
- **PitAdjacent[1,1]** $\Rightarrow$ Pit is in one of the neighbors of [1,1]: candidate set = {[1,2], [2,1]}
- Since the agent has stood at [1,1] and **no Stench was perceived there**, by the contrapositive of Rule 1 ($\neg \text{Stench}[1,1] \rightarrow \neg \text{WumpusAdjacent}[1,1]$) **no Wumpus can be adjacent to [1,1]** — this eliminates [1,2] from the Wumpus candidate set (as [1,2] is a neighbor of [1,1]). $\rightarrow$ Wumpus candidates reduce to {[1,3], [2,2]}.
- Since Glitter (Gold) was safely perceived at [2,2], and a cell containing the Wumpus is normally treated as lethal/unreachable, [2,2] is inferred **not** to contain the Wumpus. $\rightarrow$ **Wumpus is most likely at [1,3]**.
- Pit candidates remain {[1,2], [2,1]} — no additional percept yet distinguishes between them without visiting one of these cells.

(c) Safety of Path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] — 3 marks

- **[1,2]:** No Breeze was reported here (only Stench), so by Rule 2's contrapositive, $\neg \text{Pit}[1,2]$ can be inferred (no pit adjacent, and specifically not at 1,2 itself under the simplifying single-hazard reading). Since Wumpus was eliminated as being *at* [1,2] itself, Rule 4 gives **Safe[1,2]**.
- **[2,2]:** No Stench or Breeze reported there, and Glitter/Gold was found (a cell is normally only reachable/observable if the agent survives entering it) $\rightarrow \neg \text{Wumpus}[2,2] \wedge \neg \text{Pit}[2,2] \rightarrow$ by Rule 4, **Safe[2,2]**.
- **Conclusion:** Using the knowledge derived in (a) and (b), the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] **is SAFE** for the agent to traverse and collect the gold, since Rule 4 evaluates to true for every cell on the route.

Summary of Marks

Case Study	(a)	(b)	(c)	Total
1. Medical Diagnosis	3	4	3	10
2. Traffic Management	3	4	3	10
3.	3	4	3	10

Case Study	(a)	(b)	(c)	Total
Agricultural Reasoning				
4. Wumpus World	3	4	3	10
Grand Total				40